Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/18/2006
PI Name:	Limoli, Charles Ph.D.		
Project Title:	High LET Radiation And Neurogenesis: Implications And Mechanisms Underlying Cognitive Impairment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	climoli@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-3053
Organization Name:	University of California		
PI Address 1:	Dept. of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
Start Date:	09/01/2003	**End Date:**	05/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	6/16/2006--PI transferring project to new institution; closing out UCSF grant.		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1632		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	For those engaged in space travel, the radiation environment poses a range of potential health risks. For instance, it is possible that exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can induce a range of CNS injuries, including cognitive impairment. While the pathogenesis of radiation-induced cognitive impairment is not clear, we have shown that neuronal precursor cells in the hippocampus may be involved. Using low doses of X-rays (1-5 Gy), we have demonstrated, in vitro and in vivo, that neural stem cells of the hippocampus are exquisitely sensitive to irradiation. In addition, these cells exhibit oxidative stress and undergo apoptosis after irradiation. Low dose x-irradiation of these cells in vitro also leads to a persistent oxidative stress that coincides temporally with a dose-dependent reduction in neurogenesis observed in vivo. Consequently, we hypothesize that oxidative stress plays a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo. Further, we suggest that oxidative stress constitutes a biochemical mechanism that can influence the fate of stem cell differentiation in vitro and hippocampal neurogenesis in vivo. It is our overall objective to determine how oxidative stress affects the high LET radiation response of precursor cells in the subgranular zone (SGZ) of the hippocampus and to identify compounds/strategies that will enable us to ameliorate the adverse effects of high LET irradiation on hippocampal neurogenesis. We have defined 4 specific aims to address our objective. Aim 1: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to acute (days) oxidative stress and altered neurogenesis after exposure to high-energy protons. Aim 2: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to chronic (months) oxidative stress and altered differentiation/neurogenesis following exposure to high-energy iron ions. Aim 3: Determine if antioxidant treatment with SOD/catalase mimetic agents reduce high-LET induced radiation injury to neural stem cells in vitro and ameliorate the radiation-induced inhibition of neurogenesis in vivo. Aim 4: Quantify the effects of high-energy iron ions on cognitive function in wild type mice and determine if antioxidant treatment ameliorates those effects. Determine if or how oxidative stress impacts the radioresponse of neural precursor cell function. Elucidating oxidative stress as a biochemical mechanism regulating the behavior of neural precursor cells will provide significant insight into how irradiation might lead to impaired cognition during spaceflight. These studies will also provide information relevant to the development of antioxidant strategies to ameliorate the adverse effects of irradiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will help define the risks associated with radiation exposure in space. By understanding these risks NASA can improve its strategies for promoting the health of astronauts.
Task Progress:	Progress over the first three years of our funding period has been significant, as evidenced by 8 published papers. To date we have undertaken 4 runs at BNL (two in 2005 and two in 2006) as well as two runs at Loma Linda. We have now successfully irradiated a large number of in vitro samples as well as our SOD3 knockout mice, mice heterozygous for all three SOD isoforms (CuZnSOD, MnSOD and SOD3) and all relevant WT control mice. With these studies, we continue to make significant progress in completing the objectives outlined in this grant. Major accomplishments are summarized below: 1) In vitro studies using hippocampal and subventricular neural precursor cells have revealed that 56Fe-ions induce marked levels of oxidative stress. Reactive oxygen species detected after heavy ion irradiation are 2-10 fold higher at various times (6-48h) after irradiation. 2) In vitro studies using hippocampal neural precursor cells have shown protons to be more efficient at eliciting oxidative stress than X-rays. 3) The antioxidant lipoic acid has shown promise at reducing radiation-induced ROS, whether added as a radioprotector (before irradiation) or as a mitigating agent (after irradiation). 4) Comprehensive analysis of our EC-SOD (extracellular SOD) deficient (nullizygous) mice has now shown that these animals are actually protected from the inhibition of neurogenesis after irradiation. While this result is qualitatively similar between low and high LET radiations, heavy ions do show important differences, possibly due to changing levels of activated microglia. This provocative data now shows for the first time that altering the redox status of the microenvironment can be protective to certain adverse radiation effects in the animal. These studies will be expanded in the next year of funding. 5) Chronic studies analyzing oxidative stress in neural precursor cells that have been exposed to a range of Fe ion doses (0.1 – 2 Gy) have nearly been completed. This new data has revealed significant increases in oxidative stress that persist over background levels for over two months after initial irradiation. 6) Studies exposing neural precursor cells to Fe ions at very low doses (1-15 cGy) have shown oxidative stress to be significantly elevated over acute times (12-24h) after irradiation. 7) SOD1 and SOD2 heterozygous mice have all been irradiated with a range of iron ion doses. These mice have also all been sectioned and processed for immunohistochemical staining and are currently being analyzed for neurogenic changes by confocal microscopy. We expect this to be completed within the next year. 8) Funding from this Grant has resulted in 8 published manuscripts and several in preparation.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Rola R, Sarkissian V, Obenaus A, Nelson GA, Otsuka S, Limoli CL, Fike JR. "High LET irradiation induced inflammation and persistent changes in markers of hippocampal neurogenesis." Radiat Res. 2005 Oct;164(4 Pt 2):556-60. PMID: 16187787 , Oct-2005
Articles in Peer-reviewed Journals	Giedzinski E, Rola R, Fike JR, Limoli CL. "Efficient production of reactive oxygen species in neural precursor cells following exposure to 250 MeV protons." Radiat Res. 2005 Oct;164(4 Pt 2):540-4. PMID: 16187784 , Oct-2005
Articles in Peer-reviewed Journals	Limoli CL, Giedzinski E, Baure J, Rola R, Fike JR. "Altered growth and radiosensitivity in neural precursor cells subjected to oxidative stress." Int J Radiat Biol. 2006 Sep;82(9):640-7. PMID: 17050476 , Sep-2006

Articles in Peer-reviewed Journals	Limoli CL, Giedzinski E, Baure J, Doctrow SR, Rola R, Fike JR. "Using superoxide dismutase catalase mimetics to manipulate the redox environment of neural precursor cells. " Radiat Prot Dosimetry. 2006 Dec 13; [Epub ahead of print] PMID: 17166877 , Dec-2006
Articles in Peer-reviewed Journals	Limoli CL, Rola R, Giedzinski E, Mantha S, Huang TT, Fike JR. "Cell-density-dependent regulation of neural precursor cell function. " Proc Natl Acad Sci U S A. 2004 Nov 9;101(45):16052-7. PMID: 15522966 , Nov-2004